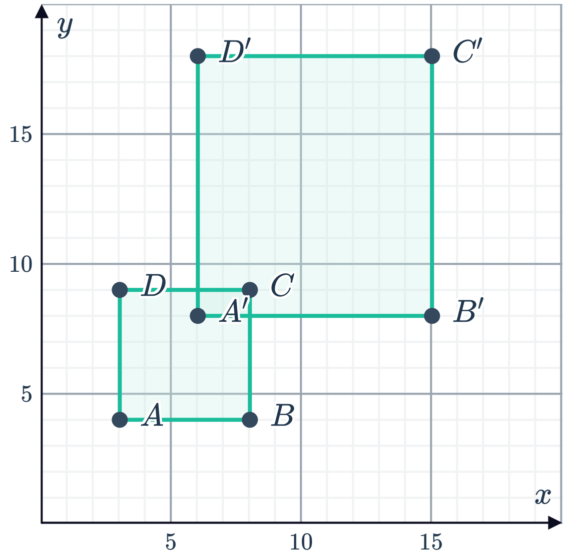
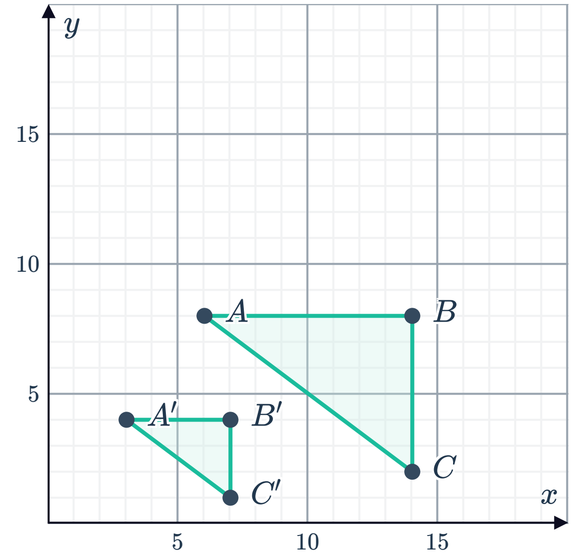
Dilations

1 Determine whether the following graphs show a dilation. If yes, find the scale factor.

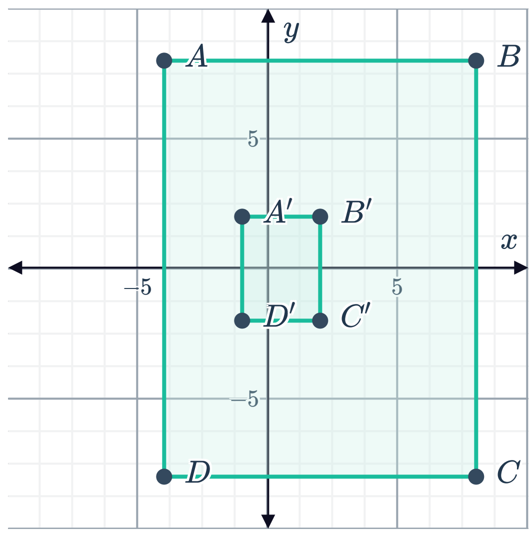
a



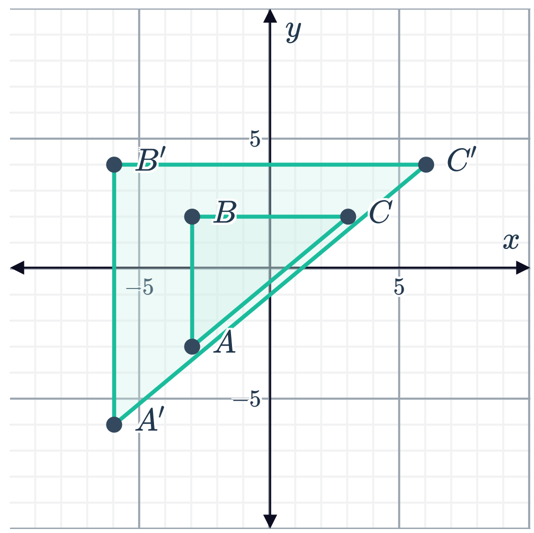
b



c

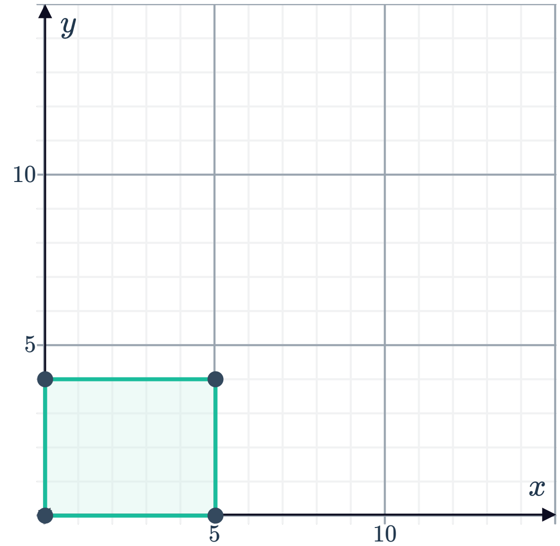


d

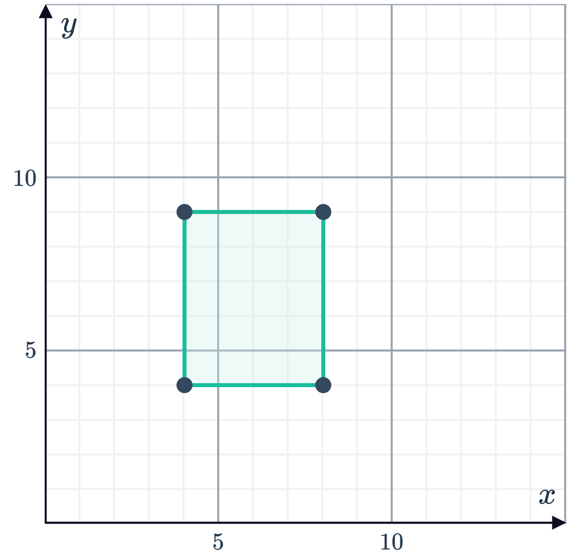


2 Dilate each figure using the given scale factor and the origin as the center of dilation.

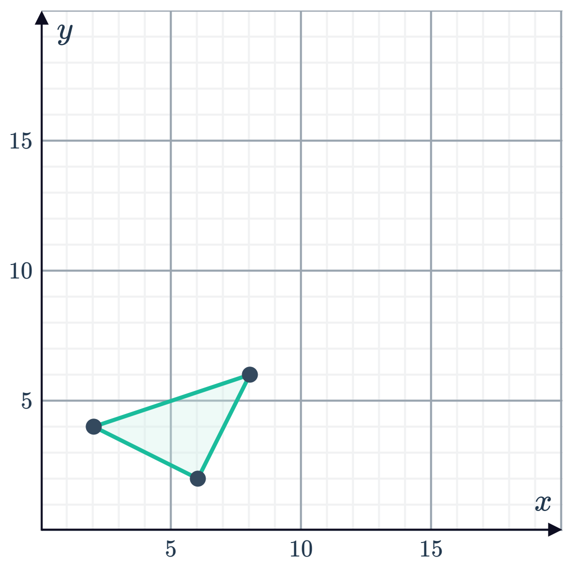
a Scale factor: 4



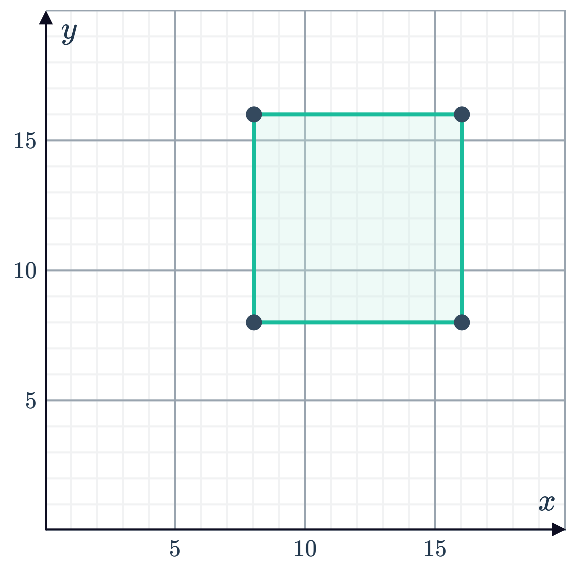
b Scale factor: 2



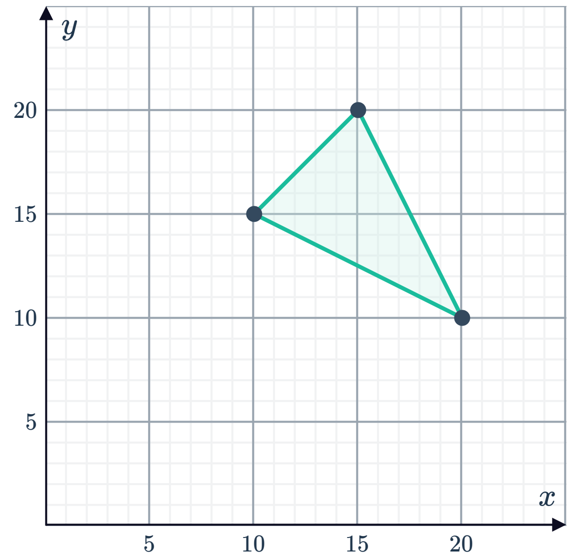
c Scale factor: 1.5



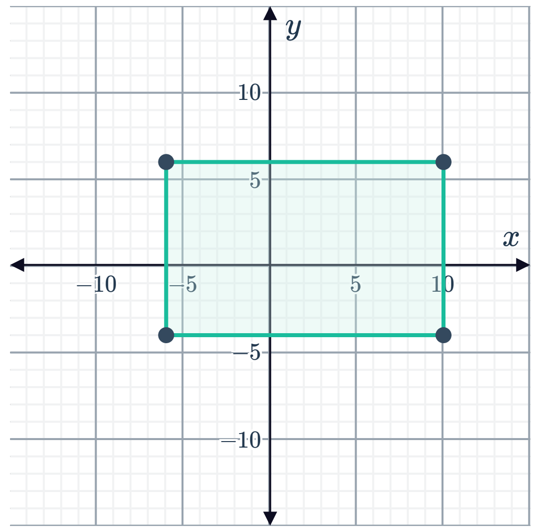
d Scale factor: $\frac{1}{2}$



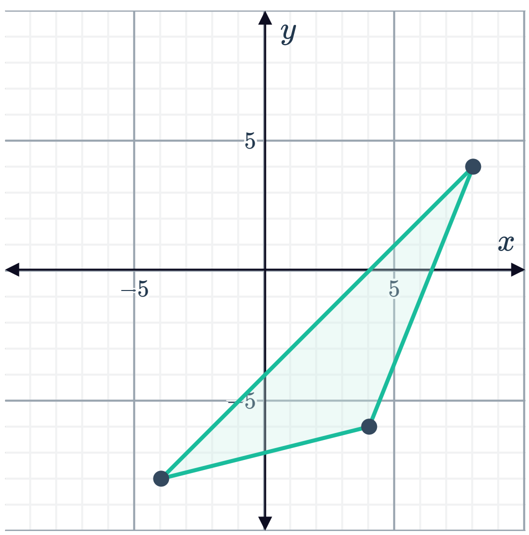
e Scale factor: $\frac{4}{5}$



Scale factor: $\frac{1}{2}$

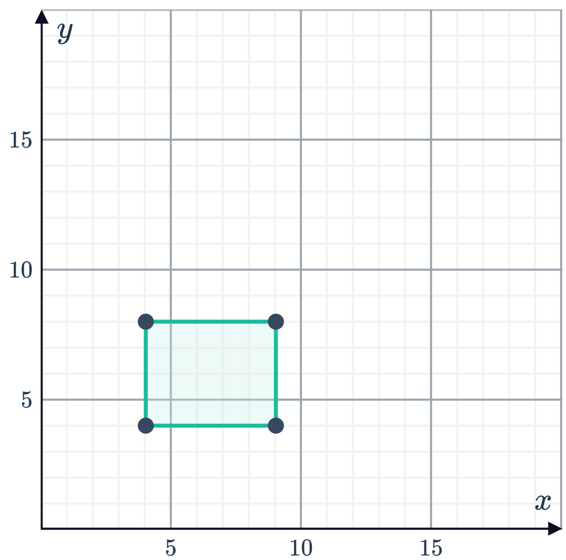


g Scale factor: $\frac{1}{2}$

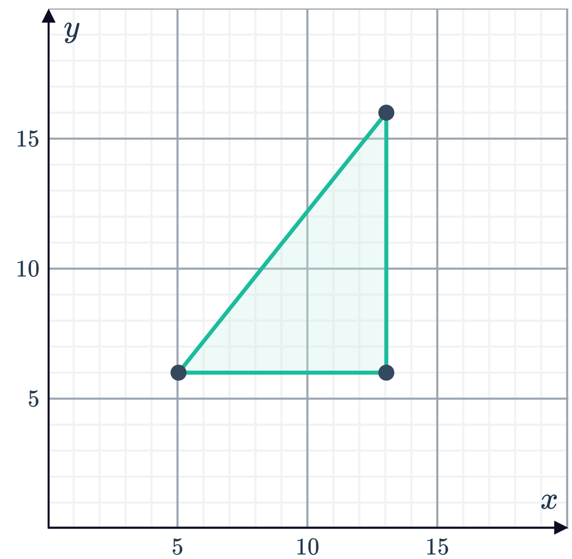


3 Dilate each figure using the given scale factor and center of dilation.

a Scale factor: 2
Center of dilation: $A\left(2,3\right)$



b Scale factor: $\frac{1}{2}$
Center of dilation: $A\left(3,2\right)$



4 A triangle with vertices $A\left(-7,-5\right)$, $B\left(3,-5\right)$ and $C\left(-5,9\right)$ is dilated by a factor of 0.5 using the origin as the center of dilation. State the coordinates of the vertices of the dilated triangle.

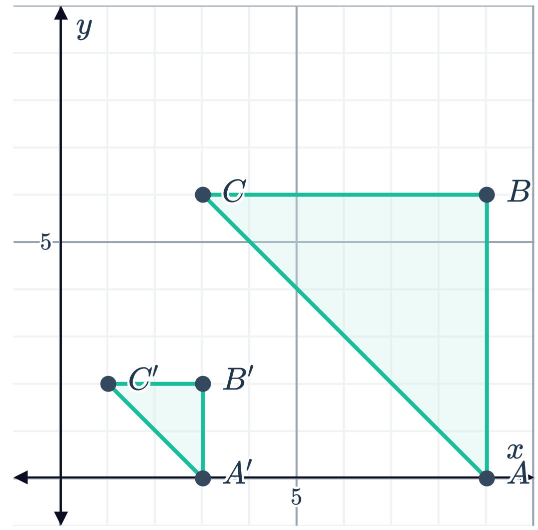
5 If a square with an area of 25 ft^2 is dilated by a factor of 0.4, find the side length of the dilated square.

6 Find the scale factor of the following dilations:

a A rectangle with vertices $A\left(-5,3\right)$, $B\left(3,3\right)$, $C\left(3,-5\right)$ and $D\left(-5,-5\right)$ is dilated using the origin as the center of dilation. The vertices of the new rectangle are $A'\left(-25,15\right)$, $B'\left(15,15\right)$, $C'\left(15,-25\right)$ and $D'\left(-25,-25\right)$.

b A rectangle with vertices $A\left(-6,9\right)$, $B\left(9,9\right)$, $C\left(9,-12\right)$ and $D\left(-6,-12\right)$ is dilated using the origin as the center of dilation. The vertices of the new rectangle are $A'\left(-4,6\right)$, $B'\left(6,6\right)$, $C'\left(6,-8\right)$ and $D'\left(-4,-8\right)$.

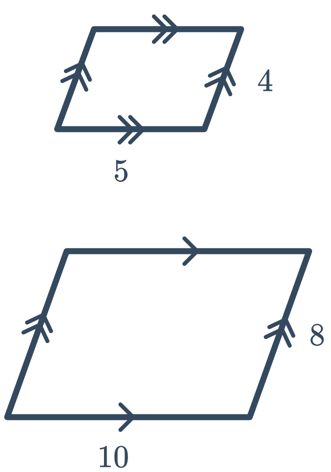
7 $\triangle ABC$ is dilated by a scale factor with a centre of dilation at the origin to obtain triangle $A'B'C'$.
Find the scale factor.



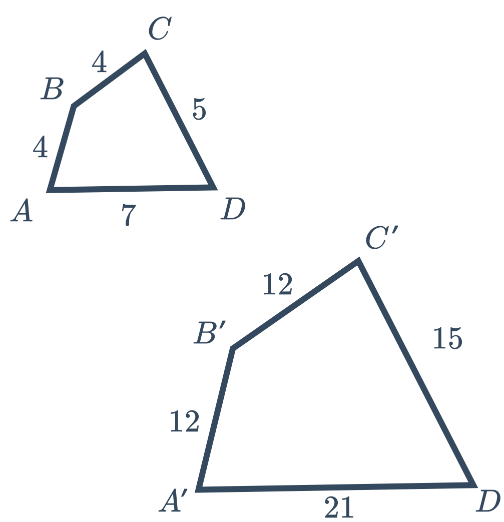
Additional questions

8 Find the scale factor for each pair of similar figures:

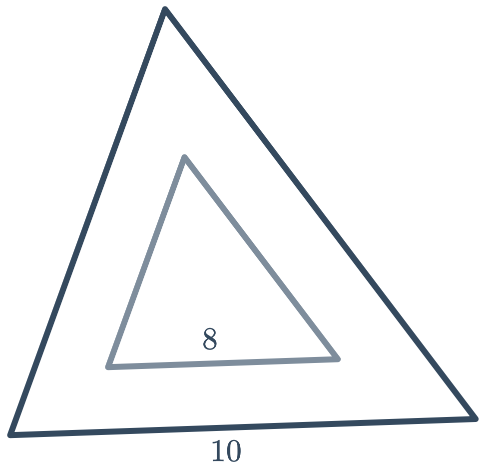
a

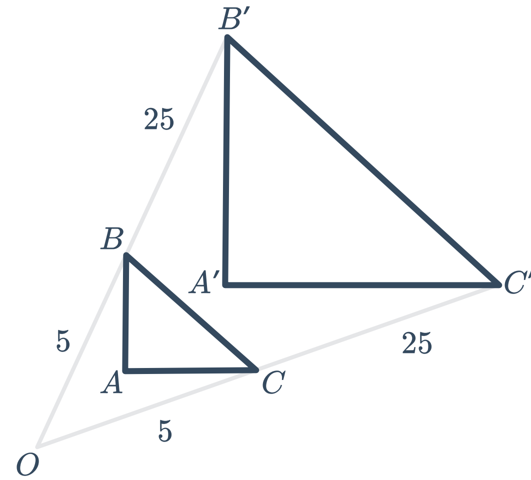


b



c





9 Determine whether each statement is true or false. Justify your answer.

a Any dilation with a scale factor less than 1 is a reduction.

b Any dilation with a scale factor greater than 1 is an enlargement.

c After dilating a figure, all corresponding angles will be congruent.

10 Layla describes the scale factor of a reduction as being in the form of $\frac{1}{k}$ since this is always a fraction less than one. Explain the error in her reasoning.

11 Use dilations to create a target: a bullseye made with four concentric circles where the width between rings is congruent. Create a diagram and justify your design.

12 Yousef needs to fit a **full** single sided sheet of notes on a single 3×5-inch notecard for his science test. Yousef's original notes are written on a standard 8.5×11-inch sheet of paper.

a Determine the largest scale factor Yousef can use to make sure the original notes fit on the notecard.

b Find the dimensions of Yousef's notes after applying the scale factor.

c Yousef realizes he can't read any of his notes because the font is smaller than $\frac{1}{12}$ of an inch after dilation. Describe how Yousef can divide and dilate his sheet of paper to utilize both sides of his notecard and still have readable notes.